

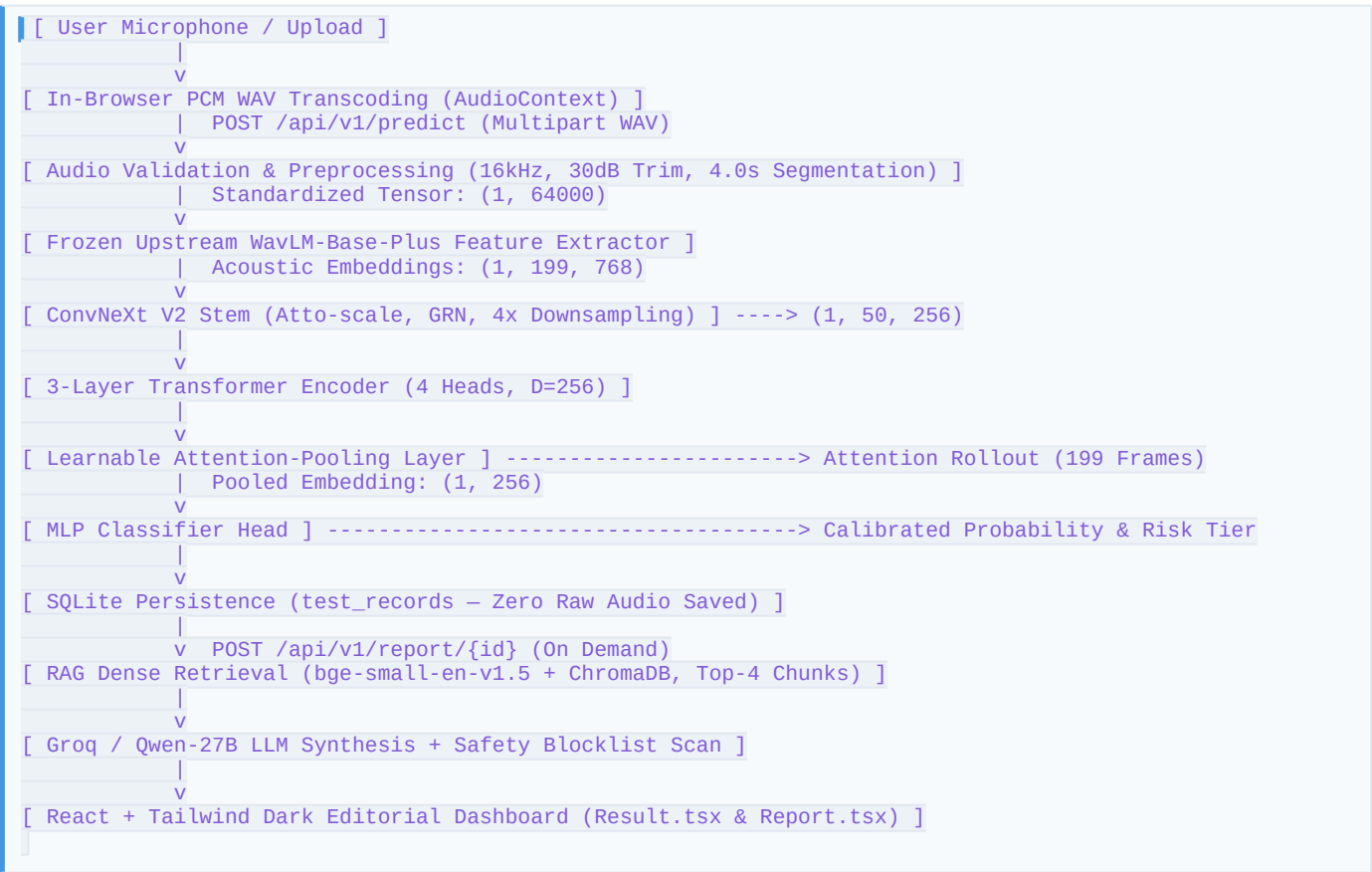
Parkinson's Disease Voice Screening Platform

Executive One-Page Technical Summary

Overview

The **Parkinson's Disease Voice Screening & Clinical Decision Support Platform (Vocalis)** is an end-to-end, privacy-preserving web application designed for non-invasive acoustic biomarker screening. Built on a hybrid deep learning architecture, the platform extracts high-dimensional self-supervised speech representations from sustained phonation audio using a frozen **WavLM-Base-Plus** foundation model, processes temporal dynamics through a custom **ConvNeXt V2 stem and Transformer Encoder** (6.42M trainable parameters), and computes millisecond-level temporal explainability via **Attention Rollout**. Predictions are synthesized into structured, non-diagnostic clinical reports using **Retrieval-Augmented Generation (RAG)** over authoritative literature stored in **ChromaDB** with defense-in-depth safety blocklists and **Groq/Qwen** LLM inference.

End-to-End System Pipeline



Core Technology Stack

- Deep Learning & Audio:** PyTorch (TorchScript), Hugging Face Transformers ([microsoft/wavlm-base-plus](#)), Librosa, SoundFile.
- RAG & Embeddings:** ChromaDB (local persistent vector store), Sentence-Transformers ([BAAI/bge-small-en-v1.5](#), 384-dim).
- LLM Synthesis:** Groq Hosted LPU ([qwen/qwen3.8-27b](#)), Google Gemini ([gemini-2.5-flash](#)), local Ollama ([qwen2.5:3b-instruct](#)).
- Backend API:** FastAPI, Uvicorn, Pydantic v2, SQLAlchemy 2.0, SQLite ([backend/app.db](#)).
- Frontend UI:** React 18, Vite, TypeScript, Tailwind CSS, Lucide Icons, Vitest.

Headline Benchmark Performance (Held-Out Test Set)

- **ROC-AUC:** 0.9645 | **PR-AUC:** 0.9712 | **Expected Calibration Error (ECE):** 0.0533
- **Accuracy (Threshold 0.55):** 91.67% (99/108 test clips correctly classified on unseen subjects).
- **Sensitivity / Recall:** 92.19% (59/64 Parkinson's phonations detected).
- **Specificity:** 90.91% (40/44 Healthy control phonations correctly identified).
- **Inference Latency:** < 250 ms on CPU for complete acoustic feature extraction and classification.

Key Technical Limitations & Safety Guardrails

1. **Screening Proxy, Not Diagnosis:** Evaluates vocal acoustic stability only; cannot replace physical MDS-UPDRS motor exams, neurological history, or DaTscan imaging.
2. **Linguistic & Demographic Scope:** Primary training data consists of Italian speakers (IPVS, 65 subjects); cross-dataset English continuous reading (MDVR-KCL) shows domain shift (ROC-AUC 0.4554).
3. **Defense-in-Depth Safety:** Multi-tiered protection—system prompt mandates non-diagnostic language, a post-generation blocklist discards forbidden assertions ("you have parkinson", "diagnosed with"), and the legal disclaimer is hardcoded as an immutable Python constant.
4. **Zero Audio Persistence:** Raw vocal audio waveforms are held exclusively in volatile RAM during the HTTP request lifecycle and are never written to disk or database.